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CIS 4394 Agentic AI

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Individual Project Proposal: The Planner Agent

Problem Statement & Motivation

Travel planning requires balancing constraints such as budget, time, transportation, and personal preferences, often across multiple platforms. When constraints such as budget reductions or flight delays change, travelers must rework their plans. Existing travel tools focus on booking and lack flexibility or transparency in planning decisions.

This project addresses the need for adaptive, personalized travel planning for students, young professionals, and budget-conscious travelers. An agentic AI solution is appropriate because travel planning involves interdependent tasks that would benefit from autonomous reasoning. A multi-agent system can dynamically revise itineraries, negotiate trade-offs, and clearly explain decisions more effectively than a static workflow.

Use Cases & Scenarios

Scenario 1: Budget-Constrained Trip Planning

A user wants to plan a five-day trip to New York City with a maximum budget of $1,200.

The system collects trip dates, preferences, and budget constraints. The Planner Agent generates an initial itinerary, while the Budget Optimizer Agent evaluates costs for stays, transportation, and activities. The Local Attraction Agent recommends attractions aligned with the user's interests, and the Critic Agent reviews the plan for feasibility and consistency before presenting it to the user.

Scenario 2: Dynamic Replanning After Constraint Changes

A user already has an itinerary but loses a day due to a flight delay.

The Planner Agent revises the itinerary based on the updated constraints. The Budget Optimizer Agent renegotiates trade-offs, such as replacing paid attractions with free options. The Critic Agent ensures the revised plan still aligns with the user's goals and explains any compromises.

Initial System Design

The system will consist of a user-facing web interface and a multi-agent backend. Users input travel constraints and preferences through a simple form, and the agents generate an itinerary.

Preliminary Agents and Roles

- **Planner Agent:** Creates the overall itinerary structure, including daily schedules and sequencing.
- **Budget Optimizer Agent:** Estimates costs and negotiates trade-offs to keep the plan within budget.
- **Local Attraction Agent:** Recommends attractions, dining, and activities based on location and preferences.
- **Critic / Reviewer Agent:** Evaluates feasibility, coherence, and alignment with user goals and requests revisions when needed.

Expected Tools, APIs, and Data Sources

- Public travel or location datasets (or mocked data for development)
- Map or distance data for proximity calculations
- Static or estimated pricing datasets for budgeting

- Local CSV files for attractions and itineraries

Feasibility & Risks

Potential challenges include outdated pricing information, over-constrained user inputs, tool failures, and long context handling for complex itineraries. Constraints include API rate limits, privacy concerns, and computational limits when running multiple agents. These risks will be mitigated by using bounded cost estimates, providing clear user feedback when constraints conflict, and modular agent design to isolate failures.

The project timeline will be aligned with the class duration:

- Weeks 1–2: Proposal and system design
- Weeks 3–4: Front-end development (Milestone I)
- Weeks 5–6: Multi-agent workflow implementation (Milestone II)
- Weeks 7–8: Evaluation plan and pilot testing (Milestone III)
- Final weeks: Refinement, final report, and presentation video

The final deliverables include a written proposal, functional front-end, multi-agent backend, evaluation report, and final implementation with presentation video.